Inorganic Cumulative Exam

September 2011

You have prepared a cobalt complex by the reaction of cobaltous acetate with a 3 equivalents of a ligand with properties you hope to use in a sensor project. The procedure was very simple; reflux the mixture for 8 hours in methanol, cool, remove the methanol by evaporation, dissolve in water and precipitate with ammonium hexafluorophosphate. Recrystalize from water/methanol. The oxidation state of the cobalt and its spin state are key components in the use of the complex.

1. Describe the possible oxidation and spin states that are most likely to occur and why.
2. Describe how UV/Vis spectroscopy can help (or not) in the assignment of the oxidation and spin state.
3. Describe how elemental analysis can help (or not) in the assignment of the oxidation and spin state.
4. Describe how NMR can help (or not) in the assignment of the oxidation and spin state.
5. Describe how ESR can help (or not) in the assignment of the oxidation and spin state.
6. Describe how magnetic susceptibility measurements can help (or not) in the assignment of the oxidation and spin state.
7. Cyclic voltammetry can help (or not) in the assignment of the oxidation and spin state.